

Voice Commerce POC

Overview

Voice-enabled commerce search platform built to improve field-sales productivity and reduce dependency on high-cost vendor solutions. Designed, implemented, and demonstrated as a working Proof of Concept.

Business Problem

- Field sales teams lose time manually searching large product catalogs on mobile devices.
 - Existing vendor platforms introduce high recurring costs and limited flexibility.
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Solution

A **voice-first, AI-powered search experience** that allows sales reps to speak natural queries and instantly receive filtered product results.

Architecture Flow:

Angular (Voice-to-Text UI) → FastAPI (NLP & Intent Processing) → SolrCloud (Search & Facets)

Data Ingestion:

Apache Spark (Scala) ETL loading CSV, JSON, and flat files into Solr.

Key Technologies

- **Frontend:** Angular, Browser Voice-to-Text APIs
 - **Backend:** FastAPI, Python
 - **NLP / AI:** spaCy, BERT (intent & entity extraction), phonetics handling
 - **Search:** Apache Solr / SolrCloud
 - **ETL:** Apache Spark (Scala)
 - **Platform:** Docker, Kubernetes (cloud or on-prem)
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What Makes This Ideal

- **Voice-first productivity:** Hands-free search for sales reps in the field
- **Context-aware search:** Intent, entities, and filters auto-applied
- **Cost-efficient:** Open-source stack, no search licensing fees
- **Scalable:** One Solr cluster supports multiple regions and product lines

- **Low IT dependency:** Automated indexing and data refresh via ETL
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Demonstrated Results

- Working end-to-end POC (not conceptual)
 - Sub-second search responses with faceting
 - Clear path to replace vendor search platforms
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Target Use Cases

- Field Sales & Outside Sales Teams
 - Inside Sales / Customer Support
 - Future extension to customer-facing eCommerce
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Demo

Demo Video:

- <https://www.youtube.com/watch?v=g3jXxjRrYSM>
- https://youtu.be/lbYHw_JpQSE
- <https://youtu.be/aPLI3b5KtZI>

 **Code / Architecture:** <https://github.com/Anandvinoth/DemoAI>

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